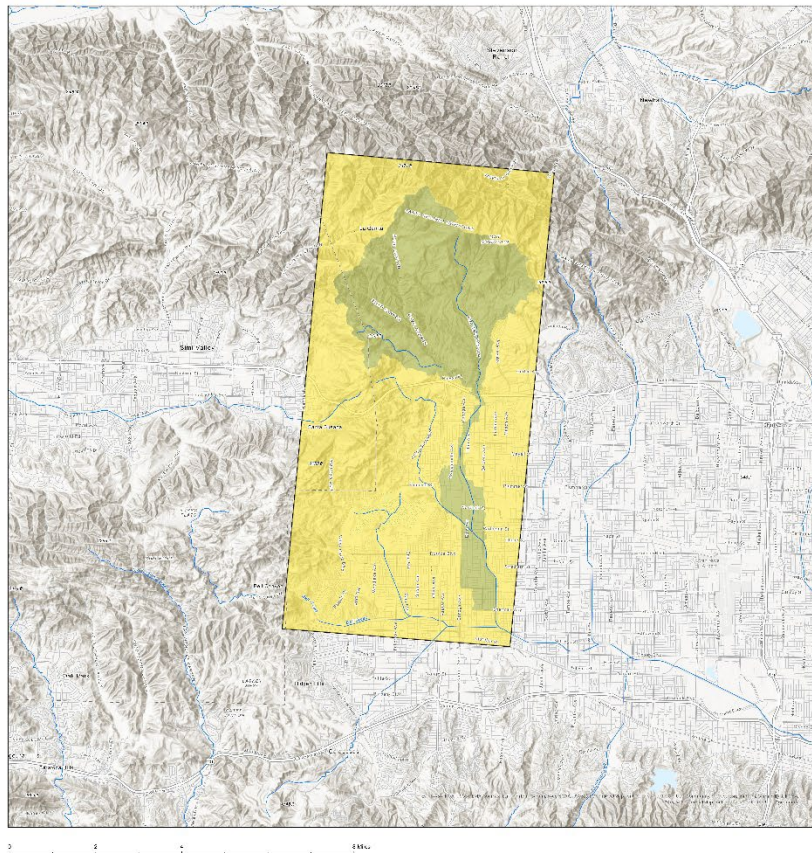


Terrain and Watershed Analysis of Browns Canyon Wash at Multiple Scales

This project explores the impact of scale on terrain and hydrologic modeling using three different resolutions of digital elevation models (DEMs): 1m, 10m, and 30m. The main objective is to compare how elevation-derived raster analyses (hillshade, slope, aspect) and watershed delineation processes vary across these scales. The project also investigates the methodological implications of working with each resolution, reflecting on accuracy, usability, and computational requirements. Building on the completed 10m DEM analysis, this report expands the workflow to include both 1m and 30m datasets. During this process several workflow issues were troubleshoot. A key challenge was with the Fill tool, which returned NoData values across the raster when applied to the 1m DEM, this issue persisted despite multiple attempts to reformat the raster using the Copy Raster tool and adjust parameters like the NoData value and pixel type but it was finally resolve after I fully updated my version of ArcGIS pro. Later in the workflow, the Watershed tool also failed to execute when using pour points, likely due to misalignment or snapping problems with the flow accumulation raster, the issue was resolved by adding multiple points instead of just one. These complications emphasized the importance of aligning input layers spatially and structurally while also be aware of the functionality of each tool before running hydrologic tools.



The Browns Canyon Study Area

Browns Canyon is located in the northwestern region of the San Fernando Valley in Los Angeles County, California. This area includes a diverse range of topography, land use types, and ecological zones. The canyon serves as a critical natural corridor, providing habitat connectivity, recreational opportunities, and potential areas for conservation and restoration.

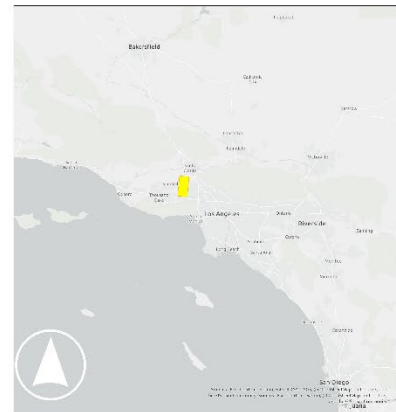


Figure 1 Study area map showing Browns Canyon Wash bounding box on World Imagery.

The study focuses on Browns Canyon Wash in the San Fernando Valley, California. The analysis area covers a bounding rectangle that includes the Browns Canyon Creek and its surrounding terrain within the Santa Susana Mountains. This region is a sub-watershed of the Los Angeles River system and experiences significant hydrologic influence from seasonal precipitation events. A base map of the area was created using the World Imagery with Labels basemap, overlaying the bounding box layer to frame the study extent. *Figure 1*

Data Table

Dataset Name	Description	Use in Analysis	Source
USGS_DEM_1m	1-meter resolution DEM	Terrain and watershed modeling	USGS National Map
USGS_DEM_10m	10-meter resolution DEM	Terrain and watershed modeling (reference layer)	USGS National Map
USGS_DEM_30m	30-meter resolution DEM	Terrain and watershed modeling	USGS National Map
World Imagery	Satellite base map	Visualization of study area	Esri Basemap

Methods

Each DEM was used to conduct identical workflows for terrain and hydrologic analysis. Terrain analysis included generating hillshade, slope, and aspect rasters using ArcGIS Pro's Spatial Analyst tools. Hydrologic modeling involved creating flow direction and flow accumulation rasters, followed by identifying pour points and delineating watersheds.

To create realistic terrain visualizations, the Hillshade tool was used with a z-factor of 1, azimuth of 315°, and altitude of 45, settings that simulate mid-afternoon lighting typical in Southern California. In addition, I experimented with various hillshade settings to simulate different lighting conditions during the day (morning, midday, and late afternoon), adjusting azimuth, altitude, and Z values to exaggerate elevation changes. This was done to visually compare the effectiveness of different illumination angles and determine which combination produced the most relevant and interpretable terrain visualization for this project. A table was created to document the combinations tested, including azimuth angles of 90°, 180°, and 315°, altitude values ranging from 30° to 75°, and Z factors from 1 to 4.

	Late Afternoon / default		Midday Light		Morning Light	
	DEM 1 (a)	DEM 30 (a)	DEM 1 (b)	DEM 30 (b)	DEM 1 (c)	DEM 30 (c)
Azimuth	315	315	180	180	90	90
Altitude:	45	45	75	75	30	30
Z value	1	1	1	2	1	4

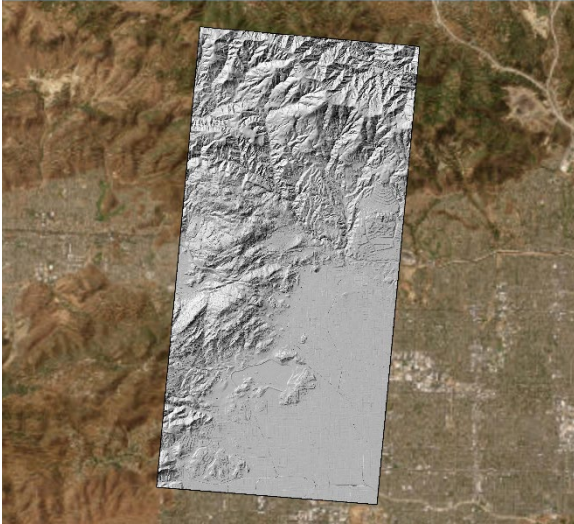


Figure 2 DEM1 (a)

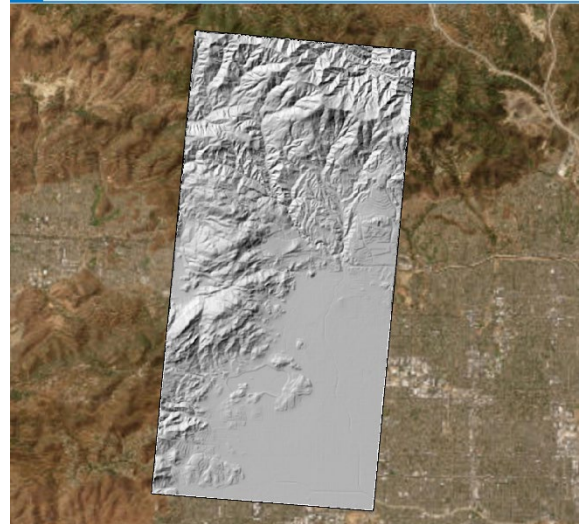


Figure 3 DEM30 (a)



Figure 5 DEM1 (b)



Figure 4 DEM30 (b)

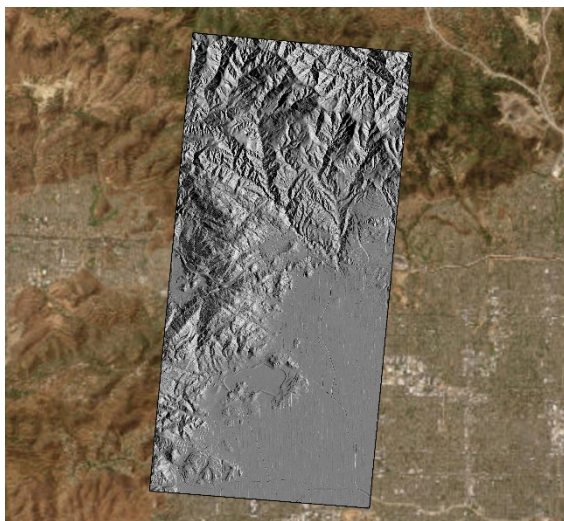


Figure 6 DEM1 (c)

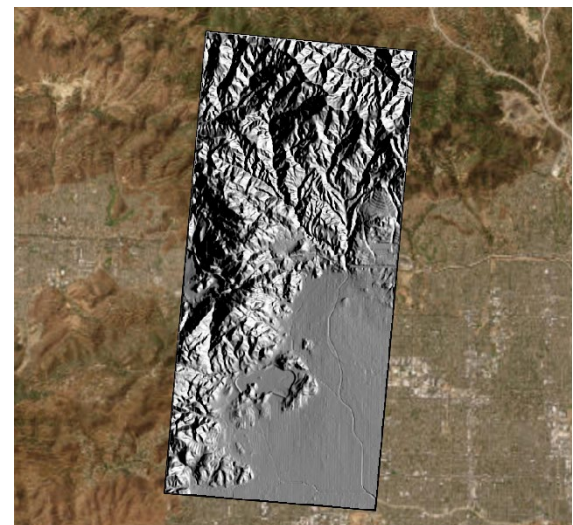


Figure 7 DEM30 (c)

For slope analysis, values were output in degrees using the planar method, while aspect was derived in azimuth degrees from north (Bolstad 2019). The Z value was change from 1 to 3 in both scales in order to visually compare.

	DEM1	DEM30	DEM1(3)	DEM30(3)
Z factor	1	1	3	3



Figure 8 DEM1(1)



Figure 9 DEM1(3)



Figure 10 DEM 30 (1)



Figure 11 DEM30 (3)

One significant issue happened with the 1m DEM during the application of the Fill tool, which produced NoData values in some parts of raster. This problem persisted despite multiple attempts to reformat the dataset using the Copy Raster tool, adjusting NoData handling and pixel type. Ultimately, the issue was resolved by updating ArcGIS Pro to the latest version, which restored normal Fill tool functionality and allowed preprocessing to continue as intended. *Figure 12*

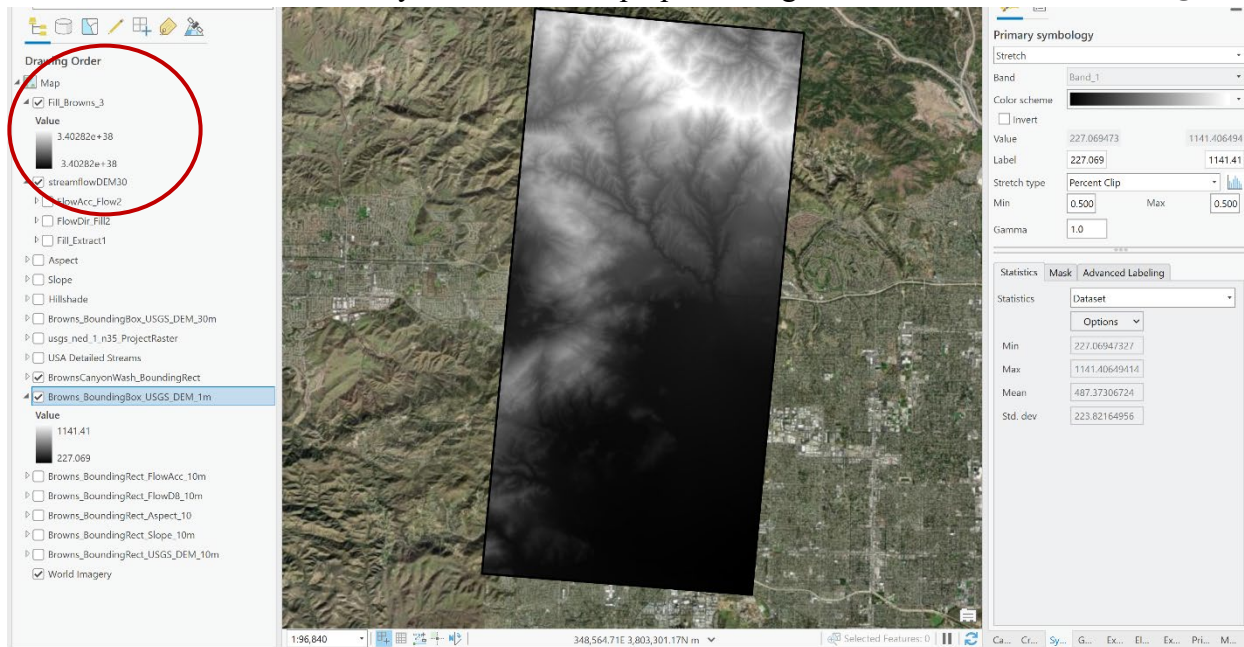


Figure 12

A second major challenge occurred when running the Watershed tool. Initial attempts using standard pour points resulted in a repeated tool failure. This issue likely stemmed from misalignment between the pour points and the flow accumulation raster, or from spatial snapping inaccuracies. To address this, multiple points were manually placed on verified high-flow cells identified through the Flow Accumulation raster (by manually I mean inside the watershed tool instead of the layer), ensuring they correspond with established drainage paths. This manual workaround enabled successful execution of the Watershed tool and proper delineation of watershed boundaries.

To determine the appropriate cutoff for flow accumulation symbology, I used a trial and error approach. I adjusted the classification values in the symbology settings of the Flow Accumulation raster to visually reduce the influence of straight line artifacts, which typically correspond to urban streets and do not represent natural hydrologic flow. This was particularly straightforward for the 30m DEM, where generalization helped suppress linear features. However, in the 1m DEM, this task was much more difficult due to the fine resolution and overwhelming detail. It required frequent zooming in and out to evaluate the pixel patterns and refine the thresholds. The higher

resolution also made subsequent watershed delineation noticeably slower and more sensitive to pixel-level variations.

For identifying the precise pixel for land area calculation and watershed initiation, I followed the suggested procedure of selecting the pixel just before Browns Canyon Creek joins the Los Angeles River. This location was determined by referencing the highest-value pixel in the Flow Accumulation raster. Using the Edit tab, I placed a point directly on the identified cell. This process worked seamlessly for the 10m and 30m DEM. However, in the case of the 1m DEM, the river alignment was noticeably off when compared with both the 10m and 30m outputs and the USGS reference shapefile. As a result, the point placement relied more heavily on visual approximation and zooming in and out to compensate for this discrepancy in alignment and ensure the pour point corresponded with the intended outlet location. *Figure 13*

Convert the pixel value to land area in km²

	pixel value	area in km ²
DEM 1m	40071388	40.071388 km ²
DEM 10m	620370	62.037km ²
DEM 30m	69697	62.7273 km ²

(PVx900)/1000000

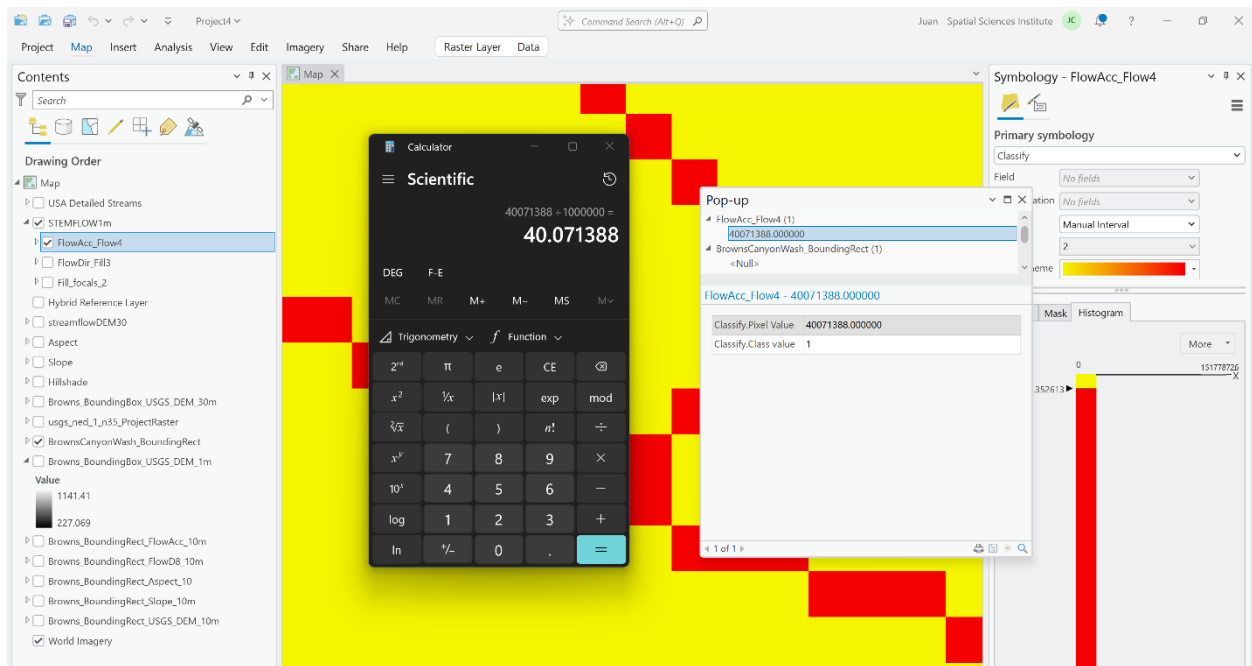


Figure 13

Results

The terrain analysis outputs, like aspect, demonstrate how scale affects the clarity and pattern of slope orientation across the Browns Canyon Wash area. The 1m aspect map captures highly fragmented terrain detail, revealing precise variations in slope direction. However, this detail can also introduce visual noise. The 10m aspect output, by contrast, strikes a balance between clarity and generalization, making it easier to distinguish dominant slope orientations across ridges and valleys. The 30m aspect map further smooths these transitions, losing some finer detail but enhancing efficiency. Overall, the 10m aspect map was the most readable for understanding terrain configuration. *Figure 14*

Watershed delineation maps provided additional insight into the relationship between DEM resolution and hydrologic modeling outcomes. The 1m watershed boundary appeared more irregular and over detailed, reflecting the high sensitivity of this resolution to small terrain variations. The 10m watershed aligned closely with both the 30m result and the USGS reference, offering consistent and reliable hydrologic boundaries. The 30m watershed map, while generalized, effectively traced the primary drainage network and main watershed extent. These results reinforce the idea that finer resolution does not always yield better hydrologic interpretation, especially when excessive detail complicates model performance and interpretation. *Figure 15*

The terrain analysis results showed that the 1m DEM produced highly detailed outputs, capturing subtle micro features like narrow ridges and small scale slope changes. However, this richness also led to visual complexity and difficulty interpreting patterns due to overcrowding, without additional smoothing. Pixel values in the terrain layers represent directional degrees for aspect (0° to 360°), slope steepness in degrees, and relative illumination for hillshade. When testing various hillshade settings, it became clear that increasing the Z value exaggerated elevation, enhancing visual relief for interpretation. Changing azimuth and altitude values allowed simulation of different lighting conditions such as morning, midday, and late afternoon which helped identify the most intuitive lighting for highlighting terrain structure.

The watershed layers showed different hydrologic patterns depending on DEM resolution. Pixel values in the flow accumulation raster represented the number of upstream cells contributing flow to a given pixel, which allowed identification of high flow areas and stream paths. In the 1m DEM, these high-value pixels were often broken or misaligned due to the resolution's sensitivity, making delineation less consistent. The 10m DEM offered clearer pathways with minimal breaks, while the 30m DEM produced smooth but generalized flow networks. Overall, the 10m DEM yielded the most consistent watershed geometry, since after all it was the reference from USGS. The 30m watershed had generalized boundaries but still captured the main drainage pattern of Browns Canyon Wash, but the symbology and representation a pixelated look that doesn't read that well.

The Browns Canyon terrain analysis (aspect)

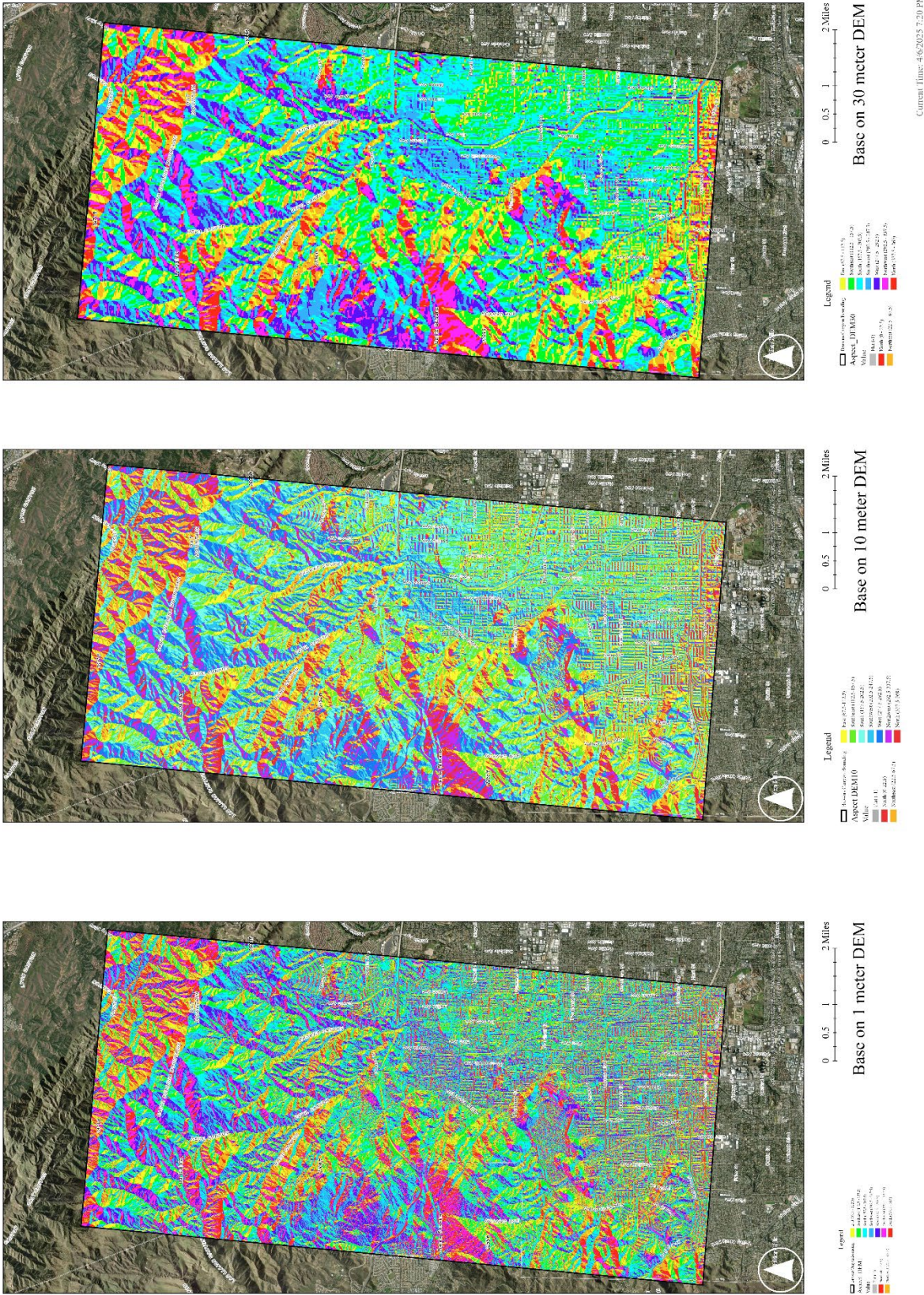


Figure 14. Aspect maps derived from the 1m, 10m, and 30m DEMs, illustrating differences in terrain orientation resolution.

the Browns Canyon Wash area

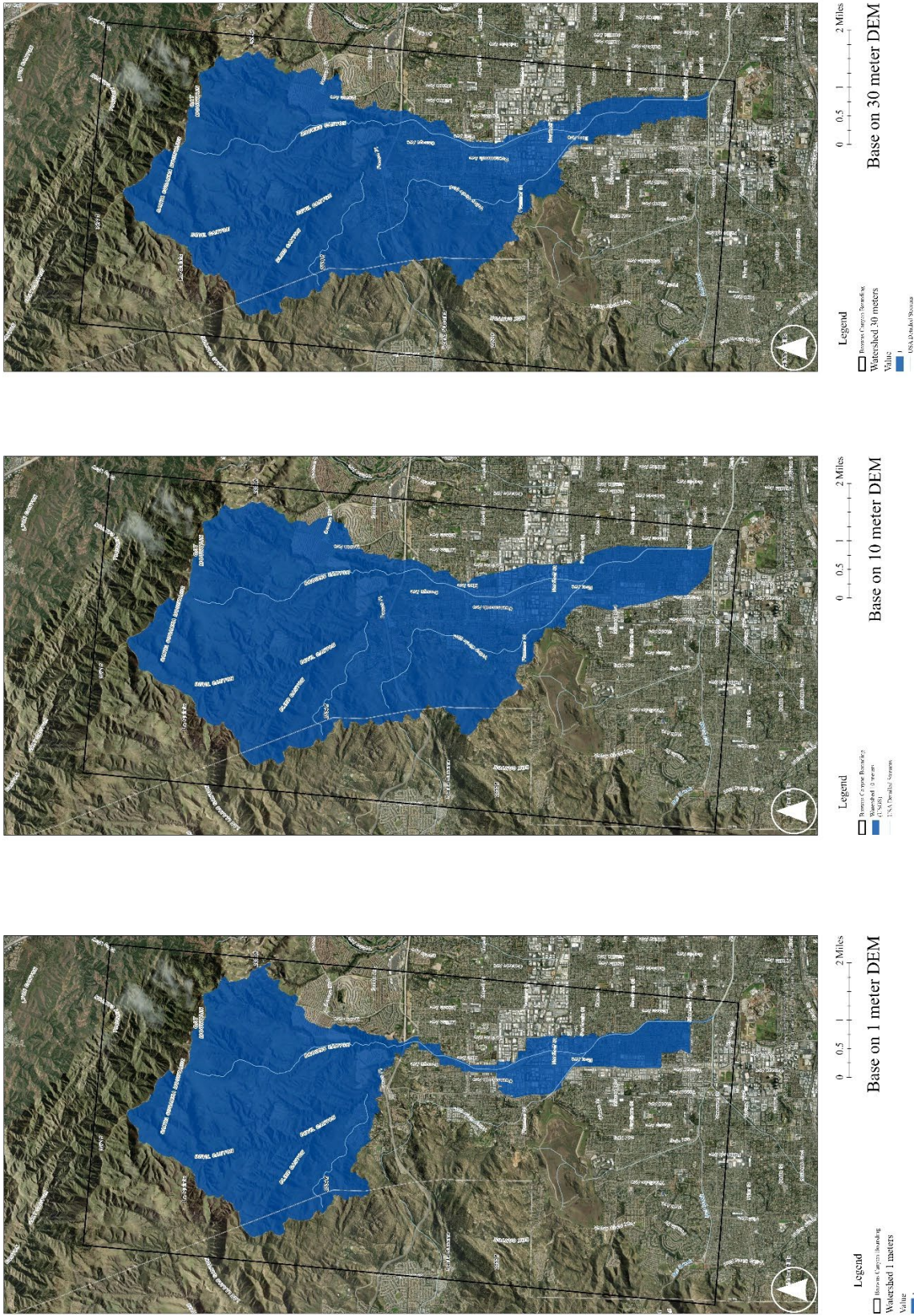


Figure 15, Watershed boundary comparisons at 1m, 10m, and 30m resolutions showing increased smoothness and alignment with official data at coarser resolutions

Discussion

Among the three DEMs, the 10m dataset stood as the most balanced choice for both terrain and hydrologic analysis. It minimized data noise while preserving relevant details, aligning well with established workflows and USGS references. The 1m DEM, while visually rich, introduced technical challenges and required significant preprocessing. It is best suited for high precision projects in smaller areas. The 30m DEM, though easy to work with, was too coarse for detailed terrain interpretation.

This comparison supports the broader observation that scale influences not just visual output but the reliability of derived products (Avelino, Baylis, and Honey-Rosés 2016). The modifiable areal unit problem (MAUP) is evident here: changes in raster resolution impacted the interpretation of watershed boundaries, particularly for the 1m data. Matching DEM scale with project goals is essential to avoid overfitting or underrepresenting landscape features (Bolstad 2019).

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